

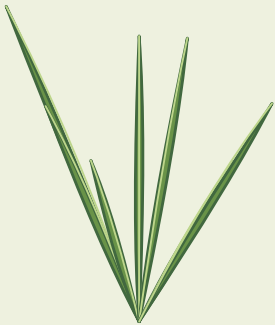
# AQUATIC PLANTS: HOW PATCHES GROW

Real growth forms  
Connected colonies  
Gentle water movement

## SALTWATER

Seagrasses are flowering plants; seaweeds are algae. A holdfast is not a root. [S1]

### Eelgrass



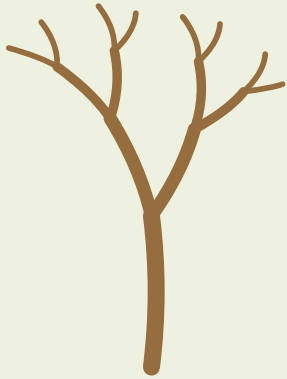
*Zostera marina* • temperate coasts [S1]  
  
Ribbon leaves arise from shoots linked by buried rhizomes. New shoots build connected meadows.

### Paddle-leaf seagrass



*Halophila ovalis* • tropical coasts [S2]  
  
Small paired leaves and creeping rhizomes create a low spreading carpet: a different silhouette from eelgrass.

### Branching brown algae



*Fucus vesiculosus* • temperate shores [S3]  
  
Tips repeatedly fork. A holdfast anchors the thallus. A branching alga is not a flowering stem.

Marine forms shown separately: *Fucus* is not a tropical reef plant.

## FRESHWATER

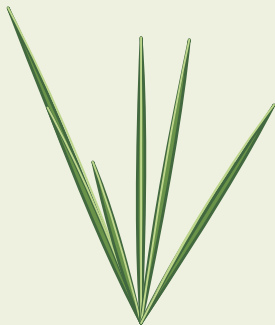
Habitat-informed references for betta and tiger barbs; regional overlap is not co-occurrence. [S7]

### Broad-leaf crypt



*Cryptocoryne cordata* • SE Asia [S4]  
  
Rhizomes and slender runners connect daughter rosettes. Variable leaves overlap in irregular patches.

### Ribbon-leaf runner



*Vallisneria spiralis* • Thailand included [S5]  
  
Stolons link new tufts. Spacer length and branching alter how widely the clone occupies the substrate.

### Feathery stem growth



*Limnophila sessiliflora* • E / SE Asia [S6]  
  
Finely divided submerged leaves form whorls. Use varied nodes and branches rather than identical paired-leaf ladders.

Betta: Thai emergent margins. Tiger barb: Sumatran freshwater. Regional flora, not one shared biotope. [S7]

## CLUMPING & BRANCHING: SMALL RULES, RICH PATTERNS

### Connected daughters



Parent-to-daughter growth makes local connected colonies. [S4, S5]

: cluster-j + 1 - ;

### Branching tips



Tip grammar:  $F \rightarrow F[+F][-F]$ . Add unequal growth, varying angles and stops. [S8]

: leaf-turn / ;

Forth inputs: two uniform values • headings in turns,  $\pm 1$  side • index/count • normalized tip height.

S1 NOAA / S2 Kew / S3 Türku / S4 Kew + Wong / S5 Kew + experiment / S6 UF

S7 habitat studies / S8 growth models / S9 flow model. Illustrative code; source links in companion plan.

### Compact vs spreading



Short spacers pack shoots; longer steps spread them. Species differ. [S8]

: fork-angle .08 \* + ;

### Root-pinned sway



Roots stay fixed; tips bend more. Neighbours share slow flow. Visual model, not CFD. [S9]

: tip-weight 0 max 1 min dup \* ;